

ARUN DAS

Full-Stack Developer · ML Engineer · MCA 2026

itsmearundasofficial@gmail.com | +91-7736908820 | Kattappana, Kerala | [LinkedIn](#) | [Github](#)

PROFESSIONAL SUMMARY

MCA student at Mar Athanasius College of Engineering with hands-on experience shipping production-grade full-stack and ML-powered web applications. Built and deployed a real-time hybrid object-detection system (YOLOv8 + EfficientNet-B0) and an end-to-end ML prediction platform with a React frontend. Proficient in Python, JavaScript, and REST API design across the MERN stack. Seeking entry-level roles in software development, ML engineering, or data analysis.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), SQL
Frontend	React.js, Hooks, Context API, Socket.io, D3.js, HTML5, CSS3, Responsive Design
Backend	Node.js, Express.js, Flask, RESTful APIs, WebSockets
ML / AI	YOLOv8, EfficientNet-B0, ByteTrack, Random Forest, scikit-learn, OpenCV, Groq API
Databases	MongoDB, MySQL
Cloud & Tools	AWS (EC2, S3, IAM), Docker, Git, GitHub, CI/CD Pipelines, VS Code

PROJECTS

Hybrid Object Detection & Classification System

Jan 2026 – Mar 2026

Python · YOLOv8n · EfficientNet-B0 · ByteTrack · Flask · React.js · OpenCV

Git: github.com/itsmearundas/object-detection

- ▶ Built a two-stage deep learning pipeline combining YOLOv8n for real-time object localisation and a fine-tuned EfficientNet-B0 classifier, achieving 85.41% test accuracy and 0.85 macro-F1 across 17 CIFAR-100 classes.
- ▶ ByteTrack multi-object tracking; Flask REST API at 43 fps (GPU) / 18 fps (CPU) with bounding-box and confidence overlays.
- ▶ Designed a responsive React frontend with mode-switching (image / video / webcam) using useEffect hooks for continuous frame streaming; real-time predictions rendered directly into the component tree - zero page refreshes.

Credit Card Default Prediction System

Jun 2025 – Dec 2025

Python · Random Forest · scikit-learn · SMOTE · React.js · Node.js · MongoDB

Git: github.com/itsmearundas/credit-default

- ▶ Trained a Random Forest classifier on 30,000 records (UCI dataset) achieving 82% accuracy and 0.79 ROC-AUC, outperforming Logistic Regression and Decision Tree baselines.
- ▶ Built an end-to-end ML pipeline: feature engineering, SMOTE for class-imbalance correction, hyperparameter tuning via GridSearchCV, and model serialisation with joblib.
- ▶ MERN platform where users input credit data → REST API → trained model → real-time default-risk predictions.

EDUCATION

Master of Computer Applications (MCA)

Aug 2024 – May 2026

Mar Athanasius College of Engineering · APJ Abdul Kalam Technological University, Kerala

CGPA: 7.75 / 10

Bachelor of Computer Applications (BCA)

Aug 2021 – Mar 2024

MES College · Mahatma Gandhi University, Kerala

CGPA: 6.36 / 10

CERTIFICATIONS & TRAINING

- ▶ Cloud Computing Workshop (5 days) - IIIT Kottayam × Educ Kshetra (Aug 2025) · AWS EC2, S3, IAM, Docker, CI/CD Pipelines, VM management.